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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS)

REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION, NOVEMBER 2025

First Semester

Discipline Specific Core course- MG1DSCBTS100

NEW BIOLOGY FOR THE NEW CENTURY

(2024 Admission Onwards)

Time: 1.5 Hours
Max. Marks: 50

Part A

Multiple Choice Questions

Answer all questions.

Each question carries 1 mark

1. (c) Dyes [K CO3]
2. (b)To regulate the progression of cell cycle.[U CO1]
3. (a) X ray crystallography [K CO3]
4. (d) All of the above [U CO2, CO3, CO7]
5. (a) Resilience [K CO4]
6. (c) Regulation of material transport [U CO1]
7. (b) They belong to different species [U CO2]
8. (a) They share the same habitat [U CO2]
9. (c) Substitution [A CO4]
10. (c) The entire DNA content present in one cell [An CO3, CO7]

Part B

Short Answer Type Questions

Answer any six questions.

Each question carries 4 marks.

11. Mass spectrometry is an analytical technique that identifies compounds by measuring the mass-to-charge ratio (m/zm/zm/z) of their ions.
    It is widely used to determine the molecular weight and structure of molecules in fields like chemistry, biology, and environmental science. (U, CO3)

12. The study of antibiotic resistance illustrates the principle of evolution by showing how bacteria adapt to survive when exposed to antibiotics. When antibiotics are used,

bacteria with random genetic mutations that make them resistant can survive and reproduce, while others die. Over time, the population shifts to mostly resistant bacteria, demonstrating natural selection—only the fittest survive and pass on their genes. This real-time evolution is a clear example of how selective pressure (like antibiotic use) leads to changes in populations, just as Charles Darwin described in his theory of evolution. (U, CO6)

13. Visualization and live imaging in disease diagnosis and therapeutics involve advanced techniques that enable real-time or detailed visualization of cells, tissues, and organs to understand disease progression and guide treatment.

- Advanced visualization technologies transform imaging data into intuitive 3D or 4D images allowing precise identification of disease sites and detailed anatomical understanding.(Techniques include CT scans, MRI, and advanced software that enhance clarity to distinguish tumors, tissue damage, or abnormalities, aiding quicker and more accurate diagnosis)

- Live imaging captures dynamic biological processes in real-time, such as cell movement, blood flow, or drug interactions, using methods like fluoroscopy, ultrasound, real-time MRI, and fluorescence microscopy. This enables monitoring of therapeutic interventions' effects at cellular or tissue levels, optimizing treatments and understanding drug mechanisms. (A, CO5,CO7)

14. Speciation is the process by which new species are formed from existing ones. It happens when populations become reproductively isolated and can no longer interbreed.Mechanisms leading to speciation include geographic isolation (allopatric speciation), reproductive isolation within the same area (sympatric speciation), and variations like peripatric and parapatric speciation. (U, CO6)

15. Independent assortment is the principle that genes for different traits separate independently of each other during the formation of gametes in meiosis. This means the inheritance of one gene does not affect the inheritance of another, leading to genetic variation. (U,CO4)

16. One limitation of Whittaker's five kingdom classification is that it places very diverse organisms together in the same kingdom. For example, unicellular algae are classified in the Protista kingdom, while multicellular algae are placed in the Plantae kingdom, even though they share similarities. This leads to inconsistency in grouping related organisms.(An, CO6)

17. The production of monoclonal antibodies involves the following steps:

1. Immunization: An animal, usually a mouse, is immunized with the target antigen to stimulate antibody production.
2. Cell Fusion: B cells from the immunized animal's spleen are fused with immortal myeloma cells to create hybridoma cells.
3. Screening: Hybridomas are screened to identify those producing the desired antibody.

4. Cloning and Expansion: Selected hybridoma cells are cloned and grown in culture to produce large amounts of the monoclonal antibody.
5. Purification: The antibodies are harvested from the culture medium and purified for use in diagnostics or therapy. (A, CO7)

18. The nucleus in eukaryotic cells functions as the control center, storing the cell's genetic material (DNA) and regulating activities like growth, protein synthesis, and cell division. It also protects DNA, oversees DNA replication and transcription, and produces ribosomes in the nucleolus.(U,CO1)

19. The genetic code is the set of rules by which the sequence of nucleotides in DNA or RNA is translated into the sequence of amino acids in a protein. Each group of three nucleotides, called a codon, codes for a specific amino acid or a stop signal during protein synthesis.(K,CO1)

20. The role of spindle fibers during cell division is to attach to chromosomes and help separate the sister chromatids or homologous chromosomes, pulling them towards opposite poles of the cell. This ensures that each daughter cell receives an equal and accurate set of chromosomes. (U,CO2)

21. Biodiversity hotspots are regions with a high variety of endemic species that exist nowhere else and have lost a significant portion of their natural habitat. Their significance lies in being critical areas for conservation because they support rich biodiversity, provide essential ecosystem services like climate regulation, pollination, and water purification, and help maintain ecological balance. Protecting these hotspots also safeguards the livelihoods of people who depend on these ecosystems.(U,CO3)

22. NMR stands for Nuclear Magnetic Resonance. Its principle is based on the fact that certain atomic nuclei have a magnetic property called spin. When placed in a strong magnetic field, these nuclei absorb and re-emit radiofrequency energy at a characteristic frequency. This resonance gives detailed information about the molecular structure and environment of the atoms in a sample.(K,CO3)

Part C

Essay Type Questions

Answer any two questions.

Each question carries 8 marks.

23. The main steps in recombinant DNA technology are:

1. Isolation of Genetic Material: Extracting the DNA containing the gene of interest from the donor organism in a pure form.

2. Cutting of DNA: Using restriction enzymes (molecular scissors) to cut the DNA at specific sites to obtain the desired gene fragment.
3. Isolation of Desired DNA Fragment: Separating the gene of interest from the rest of the DNA.
4. Amplification of Gene: Using Polymerase Chain Reaction (PCR) to make multiple copies of the gene.
5. Ligation into Vector: Joining the gene fragment with a cloning vector (usually a plasmid), using DNA ligase to form recombinant DNA.
6. Selection and Screening: Identifying and isolating transformed cells that have successfully taken up the recombinant DNA using selectable markers (e.g., antibiotic resistance).
7. Expression and Culturing: Allowing the host cells to replicate and produce the desired protein or gene product for further use.

These steps enable the production of genetically modified organisms with desired traits for research, medicine, agriculture, and industry. (K, CO4)

24. The four main types of biomolecules are:

1. Carbohydrates: They provide energy and structural support.
2. Proteins: They function as enzymes, structural components, and in cell signaling.
3. Lipids: They store energy, form cell membranes, and act as signaling molecules.
4. Nucleic acids: They store and transmit genetic information.

(Explain the basic structures of all the four) (K,CO3)

25. Gregor Mendel proposed three main principles of inheritance:

1. Law of Dominance: In a cross between two pure-breeding parents, the offspring inherit only the dominant trait visibly, while the recessive trait is masked.
2. Law of Segregation: During gamete formation, the two alleles for each trait separate so that each gamete carries only one allele.
3. Law of Independent Assortment: Genes for different traits segregate independently during gamete formation.

These principles form the foundation of modern genetics by explaining how traits are inherited in predictable patterns and how genetic variation arises in offspring.(U,CO4)

26. Biotechnology is used for medical therapies, genetically modified crops, waste treatment, and environmental conservation. Challenges include high costs, regulatory hurdles, ethical issues, and potential risks like biosecurity concerns.

    Nanotechnology offers advancements in medicine, electronics, and environmental cleanup by manipulating materials at the nanoscale. Its challenges involve safety and toxicity risks, scaling up production, regulatory gaps, ethical concerns, and interdisciplinary collaboration.

Both fields hold great promise but require cautious management to balance benefits and risks.(A,CO6)

27. The central dogma of molecular biology explains that genetic information flows from DNA to RNA and then to proteins. DNA is first transcribed into RNA, which then serves as a template for synthesizing proteins through translation. This process is essential for gene expression and controlling cellular functions.(U,CO1)

28. Independent assortment is the principle that genes for different traits separate independently during the formation of gametes in meiosis. This means the inheritance of one gene does not affect the inheritance of another. It increases genetic diversity by creating new combinations of alleles, contributing to variation among offspring.(U,CO4)

29. Biological Databases..what are they....1 mark
    Primary, secondary and composite databases with two examples each- 1 mark
    NCBI- database of databases...1 mark
    Other databases Literature databases, OMIM, KEGG and others... 1mark
    (Name of database, the data they store and the significance of databases important)
    (U,CO 5)